

HW 1 – The Normal Distribution and Combining Normal Random Variables

Directions: For the following, show ALL WORK to arrive at your answer, including a drawing. Make sure to answer the question in a complete sentence.

1) Suppose the weight of apples produced by your local orchard follows a normal distribution with a mean of 150 grams and a standard deviation of 20 grams.

(a) What is the probability that a randomly selected apple weighs more than 175 grams?

(b) What is the probability that a random selected apple weighs between 120 and 160 grams?

(c) If a grocer wants to select the top 5% of the heaviest apples to display as premium produce, what is the minimum weight an apple must have to be selected?

2) A normal distribution has mean 700 and standard deviation 50. The probability is 0.4 that a randomly selected variable from this distribution is below x . What is x ?

3) The normal random variable X has a standard deviation of 12. We also know that $P(X > 50) = 0.90$. Find the mean μ of the distribution.

4) There are two local factories producing widgets. Factory X produces widgets with a mean weight of 200 grams and a standard deviation of 20 grams. Factory Y produces widgets with a mean weight of 180 grams and a standard deviation of 25 grams. Assume that the widget weights produced by each factory are normally distributed and independent.

Let D represent the difference in weight between a widget from Factory X and a widget from Factory Y, such that $D = X - Y$. What is the probability that a randomly selected widget from Factory X weighs more than a randomly selected widget from Factory Y?